

Kiana Shi

(857) 654-7075 | shi.jiaqi1@northeastern.edu | [LinkedIn](#) | [GitHub](#)

EDUCATION

Northeastern University

Master of Science in Data Analytics Engineering

Seattle, US
Sep 2025 – May 2027 (Expected)

Northeastern University

Bachelor of Science in Computer Science & Environmental & Sustainability Sciences

Boston, US
Sep 2020 – Aug 2025

PROFESSIONAL EXPERIENCE

Wuhan Survey & Design Institute Co., Ltd.

AI Engineer Intern

Remote, China

Apr 2026 – Sep 2026

- Engineered an **LLM-based intent router** for enterprise travel and reimbursement workflows, mapping employee requests to **6 modular Skills** through structured prompting and schema-constrained JSON outputs.
- Built a **Plan-and-Execute multi-agent orchestration layer** with dynamic Skill discovery and dependency-aware scheduling, enabling parallel execution of independent tasks and ordered execution of dependent workflows.
- Designed a **1,000-query intent evaluation suite**, improving routing accuracy from **65% to 90%+** and cutting end-to-end workflow latency by **50% (30s→15s)**.

HireBeat

Software Development Engineer Intern

Remote, US

Apr 2025 – Jul 2025

- Engineered an **LLM-driven document-to-presentation pipeline** for recruiter materials, combining PDF/PPT parsing, structured content extraction, and API-based slide editing to generate editable decks from reference templates.
- Built a **multimodal layout-understanding module** using ViT-based slide representations and hierarchical clustering to group page layouts and extract reusable schemas for template-aware generation.
- Implemented an **iterative generation-and-refinement loop** with structured edit commands, execution-feedback retries, and GPT-4o-based evaluation, improving content, design quality, and structural coherence by **13%, 29%, and 25%+** over template/rule-based baselines.

Microsoft Research Asia

Software Development Engineer Intern

Beijing, China

Jun 2023 – Aug 2023

- Built a cross-dataset facial analytics pipeline spanning FER2013 and KDEF, with class rebalancing, image augmentation, and HDF5-based data pipelines for reproducible model experimentation.
- Developed ResNet50 transfer-learning models in TensorFlow/Keras with task-specific classification heads for facial emotion and gender recognition across heterogeneous image distributions.
- Conducted controlled model and preprocessing experiments, achieving **82.3% emotion accuracy on KDEF** and **99.0% gender accuracy**, with confusion-matrix and qualitative error analysis used to identify class-level failure modes and guide model selection.

PROJECT EXPERIENCE

Software Version Migration Assistant (RAG) | [GitHub](#)

- Engineered a **version-aware RAG system** for Pydantic 1.x→2.x migration, combining structured change extraction, **semantic chunking**, and constraint-aware evidence linking.
- Built a **hybrid retrieval stack** with dense embeddings, BM25, Reciprocal Rank Fusion, and version filtering, achieving **0.958 Recall@5** for change retrieval and **0.948 Recall@5** for supporting evidence.
- Designed a **multi-level evaluation framework** with a frozen Gold Set and regression tests across multi-hop, behavioral, configuration, dependency, and ambiguous queries to detect retrieval drift.

Fraud Risk Modeling & Third-Party Signal Valuation | [GitHub](#)

- Architected a **leakage-safe fraud modeling framework**, integrating feature-provenance audits, missingness diagnostics, TreeSHAP, cross-validated model selection, and temporal holdout evaluation.
- Benchmarked Optuna-tuned XGBoost/CatBoost against TabPFN-3 and TabICLv2; **TabICLv2 achieved 0.951 ROC-AUC on the temporal holdout**.
- Operationalized model scores into **approve/review/decline risk policies** and quantified the marginal value of third-party fraud signals through **vendor ablation and cost-sensitive analysis**.

SKILLS

- Programming:** Python, SQL, Java, R
- AI & LLM Systems:** RAG, Multi-Agent Systems, AgentScope, OpenAI API, Function Calling
- Machine Learning:** XGBoost, CatBoost, Scikit-learn, TensorFlow/Keras, ResNet50, Optuna, SHAP
- Retrieval & Data:** BM25, RRF, Pandas, NumPy, PostgreSQL, MySQL, Milvus, Redis, ChromaDB
- Engineering & Visualization:** Git, pytest, Streamlit, Tableau, Plotly, Matplotlib